

Project

Document role

- This file owns project scope, fixed constraints, milestone framing, sequencing posture, and near-term priority calls.
- Keep detailed subsystem design in `docs/core/SYSTEMS.md` and final 48V electrical architecture in `docs/core/ELECTRICAL_48V_ARCHITECTURE.md`.
- Keep decision/risk/open-question state in `docs/core/TRACKING.md`, not as a second decision log here.
- Keep exact topology, fuse, conductor, and bench procedure detail in `docs/implementation/`.

Snapshot

- As-of date: 2026-07-19
- Phase: hard-mounted electrical integration, bank/shore/12V commissioning, then wet-spine installation
- Install milestone: May 7, 2026 Hiatus install is historical context; the real camper shell and integrated module fit now control physical decisions
- Mission: build a reliable off-grid camper workspace suitable for full-time professional use

Fixed constraints

- Truck platform is fixed: 2021 F-350 Regular Cab Long Bed with an aluminum pickup box/body attachment environment
- Existing truck modifications are fixed: 2.5" lift, 37" tires, 4.88 regear
- Camper platform is fixed: installed/in-hand Hiatus shell on the F-350 long bed

Objectives

- Support standard workdays without power anxiety
- Maintain reliable internet for meetings and high-value workflow tasks
- Provide ergonomic, repeatable workstation setup
- Keep system serviceable and safe in multi-climate travel use

Current execution focus (electrical closeout into wet-spine integration)

- Use [PROJECT_build_order_of_operations](#) for active sequence and [LIVE_BUILD_CHECKLIST](#) for shop actions. Earlier install-window plans are historical references unless explicitly refreshed.
- The one-piece Lonseal remains the permanent finish over the three-piece 3/4" in plywood floor. Controlled module loading has begun, and the electrical module is now hard-mounted through the floor to registered truck-bed hardpoints.
- The module fits cleanly and is stable enough for continued stationary work. Final road restraint still requires the planned tie-in to the Bench/adjacent framing so the electrical frame cannot rack.
- The MultiPlus and combined AC breaker enclosure were removed before lifting to reduce module weight. Their mounting pattern is preserved by embedded pronged/spiked T-nuts in the plywood backer, so both can be remounted from the service face without loose rear nuts.
- All three 48V batteries' 2/0 AWG positive/negative branch cables are cut, lugged, heat-shrunk, and landed at the battery-side busbars. Keep the batteries isolated until Batteries 2 and 3 are charged individually, allowed to rest, and confirmed within 0.1V before the three-way parallel connection.
- Immediate electrical sequence is MultiPlus/AC-enclosure remount -> complete/protect the 10/3 shore-inlet route -> individually charge/equalize the batteries -> parallel and commission the 15.36 kWh bank -> complete the 12V buffer branch and Orion fuse cleanup -> prove factory lights, Maxxair fan, and selected DC outlets.
- Treat the water tank as a two-stage job: workbench measurement/mockup followed by one bare-tank in-truck dry fit before any new sender/fill/vent hole is cut. Lock tank restraint and wet-spine service access before the passenger-side frame closes around it.
- Shore and water-fill penetrations stay downstream of their installed inside endpoints. Prove the complete inside route, bend radius/fall, backing, service access, and strain relief before cutting the shell/bed-side interface.
- Use registered bed rivnuts rather than plywood alone for modules needing positive retention. Electrical, battery, and full-tank loads must remain visibly restrained and serviceable.
- Electrical system status remains live-proven at the system level, but permanent shore AC-in, AC-out/GFCI, full-bank commissioning, road restraint, and alternator first-charge remain separate gates. Factory pod lighting, Maxxair fan, and initial DC outlets move into the immediate 12V proof pass; solar, permanent hot water, propane closeout, and audio remain later workstreams.
- Black-walnut Galley/Desk/Bench finish surfaces remain template-gated after final floor height and module geometry are proven.

Scope

- In scope: electrical, charging, solar, communications, interior structure/cabinetry, operations process, maintenance tracking
- Out of scope for phase 1: non-critical aesthetic upgrades and major unrelated truck mechanical projects

Requirements baseline (draft)

- Workday autonomy target: load model v5 profiles indicate 3,915 Wh/day (core_workday), 4,829 Wh/day (winter_workday), and 624 Wh/day (minimal_idle_day), including conservative future-audio allowance; actual near-term office-only use may be lower. Battery reserve floor policy is 20% minimum SOC
- Connectivity reliability target: Starlink is planned primary path; fallback strategy still TBD
- Thermal comfort and electronics protection limits: TBD
- System safety requirements: fusing, disconnects, wire protection, labeling, service access

Workbook intake (from Camper Build.xlsx , imported 2026-02-11)

- Current camper config notes indicate Hiatus base model with vertical lower frame, double back doors, additional windows, Maxxair fan, and overhead LEDs.
- Consult note indicates "Into production 3rd week of February" and that delivery timeline adherence was reported as strong.
- Cabover reference captured: approximately 41 in .
- Truck config in workbook matches fixed constants: 2021 F-350, 8 ft bed.

Scope clarifications from workbook

- In scope for Phase 1 includes office-first electrical architecture, Starlink support, cabinetry/workstation, and core plumbing.
- Truck mechanical upgrades listed as "skip for now" remain out of scope for Phase 1.
- Several passthrough decisions (solar and diesel/propane routing) are still pending and tracked in docs/core/TRACKING.md ; shore hardware is selected, with inlet/cable-support details handled during physical install.

Milestone plan (draft)

- M0: requirements baseline frozen
- M1: system architecture selected (electrical + comms + thermal)
- M2: BOM v1 complete and procurement started
- M3: camper installed / physical shell available for real measurements
- M4: subsystem bench validation complete
- M5: measured interior block envelopes and service map complete
- M6: electrical module first-live checkpoint and truck-bed hard-mount complete; shore inlet, bank/ 12V commissioning, Bench tie-in, and strain-relief closeout underway
- M7: shakedown and punch-list closure

Build sequencing baseline (as-of 2026-07-19)

- The floor-removal branch is closed and the electrical-module hard-mount gate has passed. Build sequencing now keeps the battery bay and backer service face open while the heavy electrical equipment, shore route, battery bank, and 12V junction are closed out.
- Canonical sequence: remount MultiPlus/AC enclosure -> prove and install shore inlet/AC-in path -> equalize and parallel the 3x 48V bank -> finish Orion/ 12V branch and prove lights/fan/DC outlets -> bench/driveway assemble and test tank/pump/wet spine -> hard-mount tank/plumbing/Galley -> add Bench tie-in and remaining modules/panels -> shakedown.
- Detailed owner docs:
 - [PROJECT_build_order_of_operations](#)
 - [FLOORING_subfloor_build_process](#)
 - [LIVE_BUILD_CHECKLIST](#)

Immediate next decisions

- **Electrical backer remount:** reinstall the MultiPlus and AC breaker enclosure into the embedded T-nuts, then confirm backer thread engagement, service clearance, cable support, and anti-rack tie-in geometry before energizing.
- **Orion/fuse identity:** use one final standalone F-06 20A/80V FKS/ATO fuse/holder on the Orion 48V input positive lead. Identify the physically X-marked Lynx fuse by slot rather than assuming it belongs to Orion: Slot 2 requires F-03 60A/80V for the MPPT; Slot 4 remains empty. Retain F-07 60A/80V MEGA only on Orion 12V output and F-11 100A ANL only on the 12V battery positive branch.
- **Tank port map:** owner confirms four molded ports on each end—two visibly large and two smaller—with no obvious/open top port. Measure/identify the threads and inspect for a membrane-covered top boss. Current default is upper large end port for gravity fill, highest suitable upper small end port for unrestricted vent/overflow, low north-end suction to the pump, and a separate low tank drain; freeze the assignment only after installed-orientation dry fit.

- **Electrical/battery order:** remount AC equipment and install shore AC-in first; charge Batteries 2 and 3 individually; record rested voltages; parallel only at $\leq 0.1V$ difference; then verify polarity, branch protection, SmartShunt, disconnect, and current sharing before Bench closure.
- **Penetrations:** the hard-mounted electrical endpoint now exists, so the shore inlet may proceed after the exact inside/outside route, hidden-structure clearance, backer grommet/strain relief, and sealing surfaces are proven together. Water-fill/vent still waits for tank orientation, fitting method, continuous hose fall/vent rise, and service access.
- **Restraint:** use registered rivnuts for positively retained modules and confirm the battery bench, electrical module, and full water tank are not loose or dependent on plywood alone.
- **Payload:** capture door-sticker payload and staged scale weights. Treat total payload, rear axle load, rear tire load, and left/right balance as active constraints.

Electrical-to-shakedown sequence

Purpose: finish the hard-mounted utilities and close furniture around tested systems without rebuilding access twice.

1. **Electrical backer and shore AC-in**
 - Remount the MultiPlus and combined AC breaker enclosure into the embedded T-nuts; prove the complete inside/outside shore path, then cut, bed, seal, grommet, strain-relieve, and test the L5-30/ 10/3 AC-in route.
2. **Open-access battery-bank commissioning**
 - Charge Batteries 2 and 3 individually, rest and record all three voltages, parallel only within $0.1V$, then verify branch protection, polarity, disconnect, Lynx/shunt, cable support, covers, extraction, and current sharing with the Bench open.
3. **12V commissioning**
 - Install final Orion F-06, complete the 12V buffer branch, verify F-07 / F-11, then prove factory lights, Maxxair fan, pump branch, and selected DC charging outlets.
4. **Tank, restraint, and wet-spine test/install**
 - Add the approved sender/fittings, bench/driveway fill-pressure-leak-function test the tank/pump assembly, then hard-mount its restraint and removable pump/strainer/accumulator/manifold board with wet/dry separation and low-point service access.
5. **Remaining penetrations, modules, and shakedown**
 - Prove/cut the gravity-fill/vent opening only from installed geometry; add Galley/Bench/Desk/fridge modules and the electrical anti-rack tie-in, keep panels removable until tests pass, then torque/witness-mark, drive locally, and inspect for floor damage, leaks, insert spin, abrasion, module movement, and fastener loss.

Cabinetry approach (recommended)

- Treat 80/20 as the *structure* and simple panels as the *skins*.
- Prioritize serviceability over concealment: every fuse/disconnect should be reachable without uninstalling major devices.
- Build “functional first”: once the electrical module works and survives vibration, add frosted panels + LED backlighting as a Phase 2 overlay.